

Start: 2:45 pm

- Issue with rarefaction curve
- With 3145 we keep all samples but we are only keeping 22.8% of reads -> getting rid of a lot of rare stuff
 - Number is too low to see any biological relevance
 - 8000 → Keeping 46% of reads seems to be our max?
 - Between 216 and 217 in 8000 cut off with double the amount of reads
 - Sucks that some of our reads have very low depth
- Do not need to re-train the classifier (should be fine since we trained on one nucleotide or longer)
 - Can just do the rarefaction on R as well for the assignment/final project
 - Only need to rerun the DADA2 denoising, redo the tabulating, skip the training of the classifier
- Which metrics to prioritize? (Module 13)
 - Running a few metrics to see how it affects the data
 - Alpha curve for all of the alpha metrics (shannon, observed features, etc..)
 - Cancer stages along the axis of each
 - Try all of beta metrics then pick the one that looks the “best” (pretty privilege)
 - Colour by stages and shapes by sex
- Taxonomic Differences
 - Go with at least the genus level
- Functional Analysis
 - Two options on entire metadata:
 - Take the full list and take only the top 20 (the ones that had the biggest changes from the baseline) → see if they are all related to some type of metabolism
 - Think about literature and see what metabolic process is normally affected/affects gastric cancer and then graph it and see if there is a difference
 - Generally involves both options
 - Maybe inflammatory pathways, tissue repair
 - Look at full list first and then decide which ones u think may have changed based on literature
 - Just look for carcinogenic bacteria through the literature, build a list of those and then see if those come up on our stats/work
- Deadline (end of break)
 - Talk about with others
 - Proposal doesn't require us to actually do work/stats but would make it easier
 - Have zoom meeting maybe in break to talk over it
- Find any relevant literature that used OUR dataset to see whether their findings are helpful

Aim 1: Module 13

Aim 2: Module 14

Aim 3: Module 15